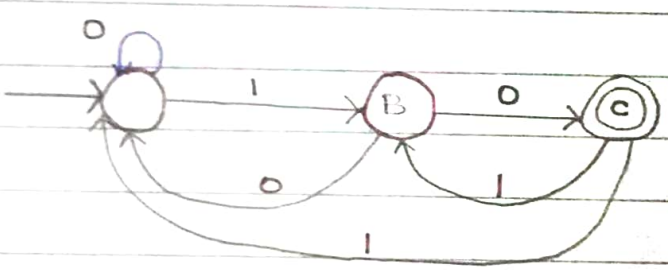
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CSA1310 - THEORY OF COMPUTATION

CO - 1

1. How can every Non - deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite automaton (DFA) and what are the challenges involved ? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0,1\}$. Construct a DFA that represents the same language L.



GIVEN:

A --0 → A (self loop)
A --1 → B
B --0 → A
B --0 → C
C --1 → B
C --1 → A

Initial state = A
Final state = C

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STATE	INPUT 0	INPUT 1
A	{A}	{B}
B	{A, C}	ϕ
C	ϕ	{A, B}

* In subset construction, each DFA state is a set of NFA states.

The DFA starts with : {A}

we will call it as q_0.

From {A}, from input 0 : $\delta(\{A\}, 0) = \{A\}$

So : $q_0 \xrightarrow{0} q_0$

For input 1 : $\delta(\{A\}, 1) = \{B\}$

Create : $q_1 = \{B\}$, So : $q_0 \xrightarrow{1} q_1$

Now process {B}

For 0, B has two possible transitions:

$B \xrightarrow{0} A$ and $B \xrightarrow{0} C$

$\therefore \delta(\{B\}, 0) = \{A, C\}$

Create : $q_2 = \{A, C\}$

PROCESS {A, B}

For 0

From A A → A

From A A → A

From B B → A, C

∴ {A} ∪ {A, C} = {A, C}

$q_3 \xrightarrow{0} q_2$

For 1:

From A : A → B

From B : nothing

(b)

∴ δ({A, B}, 1) = {B}

$q_3 \xrightarrow{1} q_1$

Dead state:

Once we reach ϕ, there are no possib
NFA states.

Therefore: $\phi \xrightarrow{0} \phi$

and $\phi \xrightarrow{1} \phi$

FINAL DFA Transition Table

DFA State	NFA state	0	1
q_0	{A}	q_0	q_1
q_1	{B}	q_2	q_d
q_2	{A, C}	q_0	q_3
q_3	{A, B}	q_2	q_1
q_d	ϕ	q_d	q_d

Initial state : $q_0 = \{A\}$

Final state : $q_2 = \{A, C\}$

2) How is mathematical induction used to prove properties of strings and language in automata theory ? Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Mathematical induction is a method used to prove that a statement is true for all positive integers. It consists mainly of two steps.

1. Base case
2. Induction step

For $n = K+1$, LHS becomes

$$1^2 + 2^2 + 3^2 + \dots + K^2 + (K+1)^2$$

Using the induction hypothesis:

$$= \frac{K(K+1)(2K+1)}{6} + (K+1)^2$$

$$(K+1)\left[\frac{K(2K+1)}{6} + (K+1)\right]$$

$$= \frac{(1+K)([K(2K+1)] + 6(K+1)]}{6}$$

$$= \frac{(K+1)[2K^2 + K + 6K + 6]}{6}$$

$$= \frac{(K+1)(2K^2 + 7K + 6)}{6}$$

$$\therefore \frac{(K+1)(K+2)(2K+3)}{6}$$

This is the required formula for:

$n = K+1$